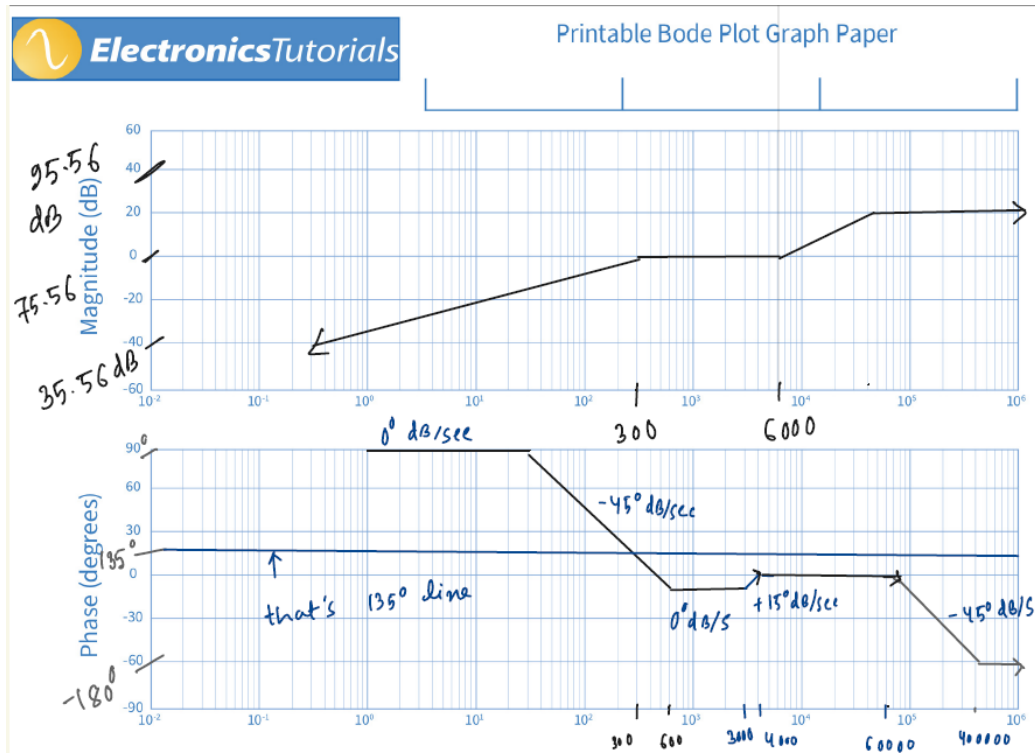


Homework 4:

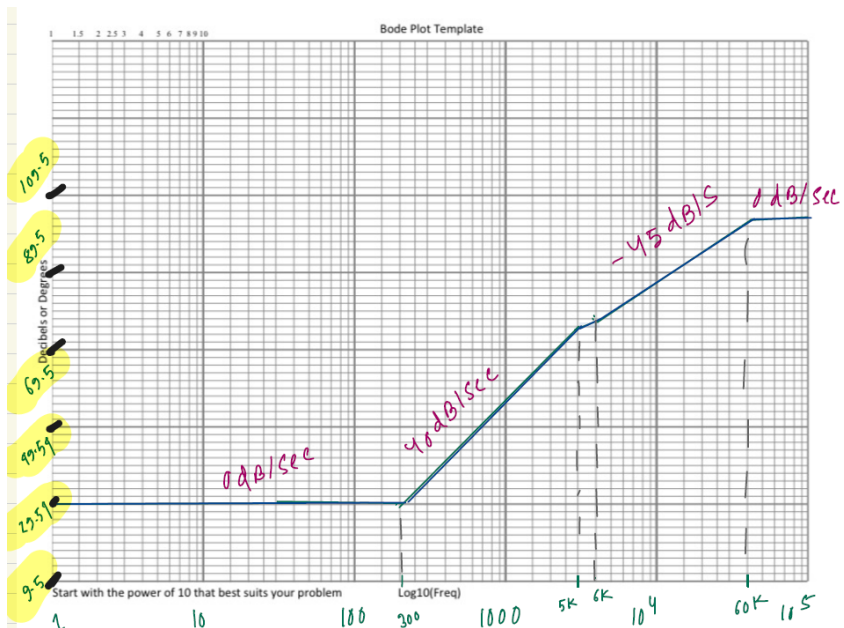
Problem 1:

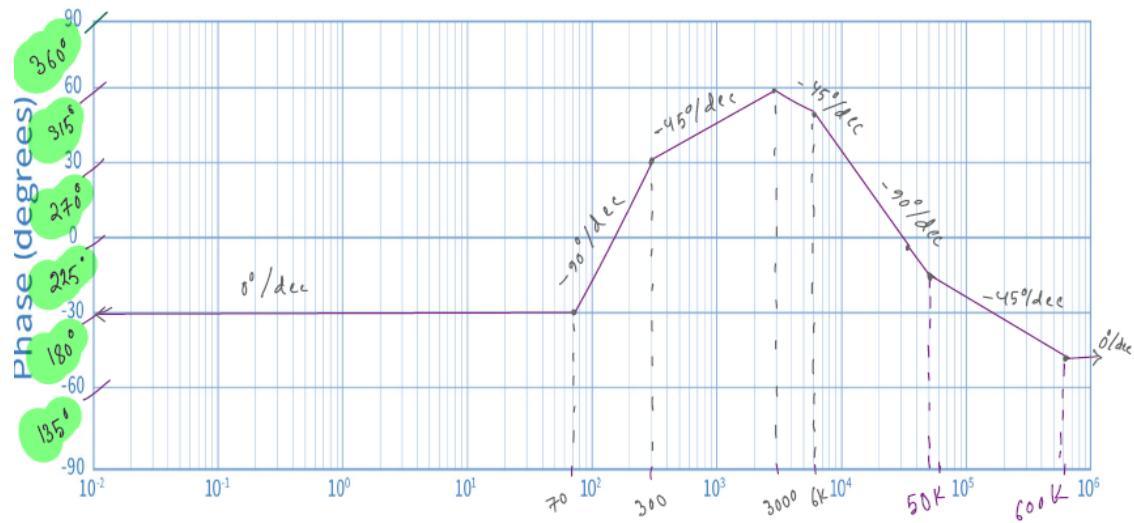
55.56 dB



Problem 2:

29.54 dB

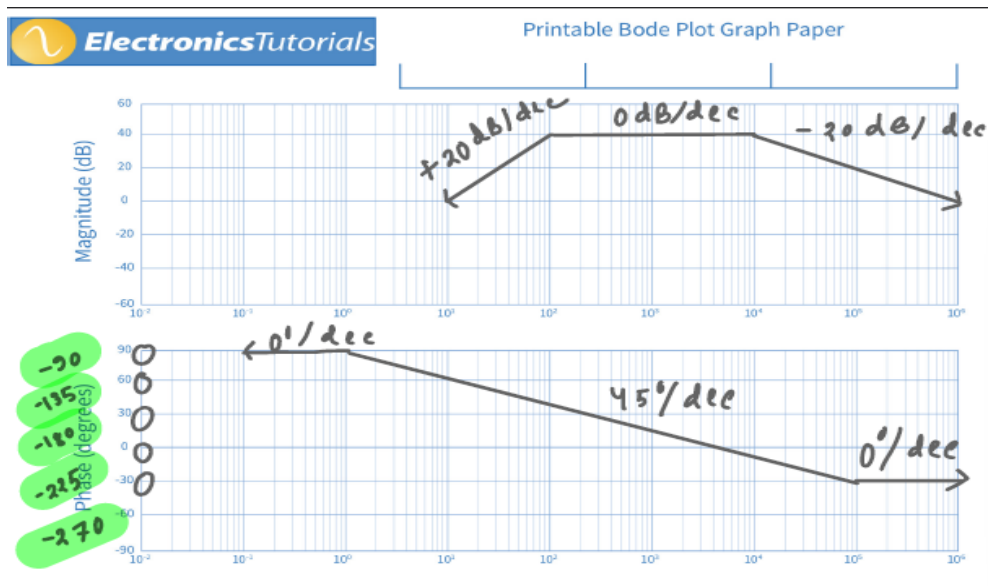




Problem 3:

$$V_{out} = \frac{GR_3K(j\omega R_4C_4)}{(1+j\omega R_4C_4)(1+j\omega R_4C_4)}$$

$$\frac{V_{out}}{V_{in}} = 20dB$$



Problem 4:

$$V_1 = V_{in} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

Problem 5:

$$i_4 = V_{in} \left(\frac{R_1 + R_2 + R_3}{R_1 R_3} \right)$$